Ashwin Ugale

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Skills Summary

AI / LLM: LangGraph, LangChain, RAG, Tool-Calling Agents, Hybrid Retrieval, Embeddings, OpenAI, Production LLM Agents, Multi-Tenant Tool-Calling, Eval Methodology (Precision@k, Labeled Sets), Human-in-the-Loop Workflows, Prompt Engineering
Backend & APIs: FastAPI, Node.js, Express, REST APIs, OAuth 2.0 + Proof-of-Possession, Microservices
Frontend: React, Redux, TypeScript, Vite, Tailwind CSS
Data: SQL, PostgreSQL, Redis, Qdrant, MongoDB
Infrastructure: AWS (EC2, S3, EKS, IAM), Azure, GCP, Docker, Kubernetes, Helm, Terraform, ArgoCD, GitHub Actions, CI/CD
Languages: Python, TypeScript, Java, Go, JavaScript, SQL

Experience

Enidus USA **August 2025 – Present**
Software Engineer NY, USA
*Building enterprise AI for **T-Mobile for Business** reseller operations.*

- Built a multi-tenant LLM tool-calling backend (FastAPI / OpenAI / Qdrant / SQL on Azure) supporting 27+ tools with JWT-enforced tenant isolation and role-based access.
- Implemented 9 LLM-orchestrated workflows (bulk operations, SIM / eSIM ordering, troubleshooting) collapsing 10–12 portal clicks per reseller action into a single chat command, with human-in-the-loop confirmation gates and SQL-persisted state.
- Designed a LangGraph-based bill-anomaly investigation agent that breaks user questions (e.g., "why did my bill go up?") into multi-step investigations across billing, usage, and equipment data, with a 50-query labeled eval set and user approval before any corrective action.
- Designed the platform's hybrid retrieval layer — Qdrant + OpenAI embeddings with LRU cache (70% hit rate) for natural-language queries, plus direct lookups for structured identifiers (phone, BAN, ICCID) to skip embedding overhead — reaching 91% top-3 precision on a 1,200-query test set.
- Migrated troubleshooting session state from in-memory dicts to a SQL-persisted state machine with typed transitions and automatic session expiration, scaling from 1 to 8 concurrent workers and preserving sessions across deploys.
- Implemented a TypeScript OAuth 2.0 + RS256 Proof-of-Possession auth layer covering ˜75% of T-Mobile's carrier API (20+ endpoints across order, billing, eligibility, search), with a translation layer mapping legacy request formats to the new API contract.
- Configured CI/CD with GitHub Actions + Azure OIDC for keyless deploys, pytest integration and retrieval-regression tests in CI, vitest, and security scanning via CodeQL, Semgrep, Dependabot, and TruffleHog.

Live Risk Labs **June 2025 – August 2025**
Software Engineer NY, USA

- Authored Terraform modules provisioning GCP infrastructure (VPCs, GKE clusters, IAM) for backend services.
- Built CIS-hardened base container images with Ansible; integrated into the CI pipeline for backend deploys.
- Set up GitOps deploys with ArgoCD; added Checkov for policy-as-code scanning on Terraform changes in CI.

Hyphenova **June 2024 – December 2024**
Software Engineer Intern California, USA

- Built reusable React + TypeScript components with Redux, lazy loading, and code splitting to improve frontend performance and modularity.
- Implemented Redis for caching, session handling, and rate limiting on a Node.js + MongoDB stack, reducing DB load by 40%.
- Configured GitHub Actions with Docker for testing, linting, and automated builds; enforced quality via Jest, ESLint, and Prettier with 85%+ test coverage.
- Integrated versioned container deployments to GKE with rollback-safe CI/CD pipelines.

Projects

- **DocChat** — version-pinned library-docs Q&A; answers from the exact version your project's lockfile pins. Built on two PyPI libraries I wrote: **smolAmem** (multi-tier agent memory — working/episodic/semantic) and **ToolPicker** (hybrid BM25 + semantic tool router). Ships as an MCP server (**docchat-server**) for Claude Code / Cursor and as a VS Code extension; 0.943 version correctness and 1.000 refusal-on-out-of-scope across a 48-pair labeled eval.

Certification

AWS Certified Developer – Associate | Link

Education

Indiana University Bloomington **August 2023 – May 2025**
Master of Science in Computer Science (GPA: 3.61/4) United States

K. J. Somaiya Institute of Technology **August 2019 – May 2023**
Bachelor of Technology in Electronics and Telecommunication Engineering (GPA: 8.99/10) India